Main Idea

- In linear algebra, there are many different kinds of vector spaces. The vectors in these spaces may be coordinate vectors, matrices, functions, polynomials, and so on.
    - Therefore, linear maps on these vector spaces may act on different kinds of vectors.
- However, thanks to their linearity, namely, their preservation of linear combinations, once we choose suitable bases, we can represent these maps by matrices.
    - Then we can study the corresponding matrices to obtain information about the original maps.
    - However, remember that the matrix representation is not the original map itself.
        - After studying the matrix representation and obtaining information from it, we should translate the results back to the original vector space.
- I think this idea is useful not only because it provides a more general way to study the properties of different linear maps, but also because coordinate matrices are much easier to manipulate in programming than vectors in their original forms.
- Here, the discussion mainly concerns finite-dimensional vector spaces, as in most problems in this course.
- In this course, the last major question is usually about an endomorphism, that is, a linear map from a vector space to itself. Therefore, the following notes mainly focus on endomorphisms.
    - However, the same general idea also applies when the domain and the codomain are different, although some topics such as eigenvalues and diagonalization are specific to endomorphisms.
- This summary was written by a first-year undergraduate student in Computer Engineering after his first Linear Algebra and Geometry exam (a bad experience). It is intended only for personal reflection and review, so the ideas and methods presented here should be treated with caution, as some parts may still be immature or incomplete.

What Properties are Preserved

- What kind of properties are preserved under the coordinate representation of the map? Or, what properties of the original map can we figure out by studying its coordinate matrix?
- Assuming linear map $f: V \longrightarrow V$, the basis of V is $\mathcal{E} = (\vec{\beta}_1, \vec{\beta}_2, \ldots, \vec{\beta}_n)$, $M = [[f(\vec{\beta}_1)]_\mathcal{E}, [f(\vec{\beta}_2)]_\mathcal{E}, \ldots, [f(\vec{\beta}_n)]_\mathcal{E}]$, $[f(\vec{v})]_\mathcal{E} = M[\vec{v}]_\mathcal{E}$
- **Key**: coordinate representation preserves linear relations(preserves linear independence)
  - $a_1\vec{v}_1 + a_2\vec{v}_2 + \cdots + a_k\vec{v}_k = \vec{0} \Longleftrightarrow a_1[\vec{v}_1]_\mathcal{E} + a_2[\vec{v}_2]_\mathcal{E} + \cdots + a_k[\vec{v}_k]_\mathcal{E} = \vec{0}$
  - Therefore, $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k$ are linearly independent iff $[\vec{v}_1]_\mathcal{E}, [\vec{v}_2]_\mathcal{E}, \ldots, [\vec{v}_k]_\mathcal{E}$ are linearly independent.
  - Thus, a basis in the coordinate space can be translated back to a basis in the original vector space.
- **Kernel Space**: $\mathrm{Ker}(f) = f^{-1}(\vec{0}) = \{\vec{x} \in V | f(\vec{x}) = \vec{0}\}$
  - The solutions of $M\vec{c} = 0$ are related to $\mathrm{Ker}(f)$:
    - $f(\vec{x}) = \vec{0} \Longleftrightarrow [f(\vec{x})]_\mathcal{E} = \vec{0}_\mathcal{E} \Longleftrightarrow M[\vec{x}]_\mathcal{E} = \vec{0}$
      - Therefore, the solutions of $M\vec{c} = 0$ are the coordinate representations of the vectors in the $\mathrm{Ker}(f)$, $[\mathrm{Ker}(f)]_\mathcal{E} = \mathrm{Ker}(M)$.
      - Namely,
        $\mathrm{Ker}(f) = \{\vec{x} \in V | f(\vec{x}) = \vec{0}\} = \{\vec{x} = (\vec{\beta}_1, \vec{\beta}_2, \ldots, \vec{\beta}_n)\vec{c} = c_1\vec{\beta}_1 + c_2\vec{\beta}_2 + \cdots + c_n\vec{\beta}_n | M\vec{c} = \vec{0}\}$
- **Image**: $\mathrm{Im}(f) = \{f(\vec{v}) | \vec{v} \in V\}$
  - How can we find $\mathrm{Im}(f)$:
    - We can find $\dim \mathrm{Im(f)}$, then find a basis of $\mathrm{Im}(f)$.
  - The image of M is related to $\mathrm{Im}(f)$:
    - $\mathrm{Im}(M) = \{M\vec{c} \mid \vec{c} \in \mathbb{R}^n\} = \mathrm{Col}(M) = \{c_1\vec{m}_1 + c_2\vec{m}_2 + \cdots + c_n\vec{m}_n \mid c_1, \ldots, c_n \in \mathbb{R}\}$, $\vec{m}_1, \ldots, \vec{m}_n$ are columns of M.
    - And $\vec{y} \in \mathrm{Im}(f) \Longleftrightarrow \vec{y} = f(\vec{v})$ for some $\vec{v} \in V \Longleftrightarrow [\vec{y}]_\mathcal{E} = [f(\vec{v})]_\mathcal{E} = M[\vec{v}]_\mathcal{E}$
      - Therefore $[\mathrm{Im}(f)]_\mathcal{E} = \mathrm{Im}(M) = \mathrm{Col}(M)$
      - Namely, $\mathrm{Im}(f) = \{(\vec{\beta}_1, \vec{\beta}_2, \ldots, \vec{\beta}_n)\vec{c} \mid \vec{c} \in [\mathrm{Im}(f)]_\mathcal{E}\} = \{(\vec{\beta}_1, \vec{\beta}_2, \ldots, \vec{\beta}_n)\vec{c} \mid \vec{c} \in \mathrm{Im}(M)\}$
      - And $\dim \mathrm{Im}(f) = \mathrm{rank}(M)$
  - So, find a basis of $\mathrm{Col}(M)$, then translate those coordinate vectors back to the original vector space, then we get a basis of $\mathrm{Im}(f)$.
- **Eigenvalues and eigenvectors**: $f(\vec{x}) = \lambda\vec{x}$, $\vec{x} \neq \vec{0}$
  - The eigenvalues and eigenvectors of M are related to those of f:
    - $f(\vec{x}) = \lambda\vec{x} \Longleftrightarrow [f(\vec{x})]_\mathcal{E} = [\lambda\vec{x}]_\mathcal{E} \Longleftrightarrow M[\vec{x}]_\mathcal{E} = \lambda[\vec{x}]_\mathcal{E}$
      - Therefore, λ is an eigenvalue of f iff λ is an eigenvalue of M.
      - And, the eigenvectors of M are the representations of the eigenvectors of f.
      - Namely, the eigenspace:
        - $E_\lambda(f) = \{\vec{x} \in V | f(\vec{x}) = \lambda\vec{x}\}$,
          - Notice, $E_\lambda(f) = \mathrm{Ker}(f - \lambda\,\mathrm{id})$ is a subspace which must include $\vec{0}$.
          - It follows that $f(\vec{x}) = \lambda\vec{x}$, $\vec{x} \neq \vec{0}$ but we cannot write
            $E_\lambda(f) = \{\vec{x} \in V, \vec{x} \neq \vec{0} | f(\vec{x}) = \lambda\vec{x}\}$.
        - $[E_\lambda(f)]_\mathcal{E} = E_\lambda(M) = \{[\vec{x}]_\mathcal{E} \in \mathbb{R}^n | M[\vec{x}]_\mathcal{E} = \lambda[\vec{x}]_\mathcal{E}\} = \mathrm{Ker}(M - \lambda I)$.
- Characteristic polynomial:
  - The characteristic polynomial $\det(M - \lambda I)$ obtained from a matrix representation of f is independent of the choice of basis.
  - Indeed, if N is the matrix representation of f with respect to another basis, then N is similar to M, so $N = P^{-1}MP$ for some invertible matrix P.
  - Hence, $\det(N - \lambda I) = \det(P^{-1}(M - \lambda I)P) = \det(M - \lambda I)$.

- This shows that the characteristic polynomial of f can be computed from any matrix representation of f.
- Therefore, we define the characteristic polynomial of f by $p_f(\lambda) := p_M(\lambda) = \det(M - \lambda I)$.

How to Solve the Last Major Question

Example: 2026/06/17 Exam Exercise

Let $A = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 2}$. Consider the map $f_A : \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ given by $f_A(B) = AB$, where AB denotes the usual matrix product.

1. Explain why f_A is an endomorphism.
2. Determine a basis of $\mathrm{Im}(f_A)$.
3. Determine a basis of $\ker(f_A)$.
4. Determine a matrix $C \in \mathbb{R}^{2\times 2}$ such that $f_A^{-1}(C) = \varnothing$.
5. Verify that $\mathrm{Spec}(f_A) = \{0, -1\}$.
6. Write the matrix associated with f_A with respect to the standard basis of $\mathbb{R}^{2\times 2}$, and write it in diagonal form if possible.
7. Determine the characteristic polynomial of f_A.

The Logic Chain in Solving the Last Major Question

- Choose or use the given basis, and represent the linear map by a coordinate matrix M.
- Study M to answer questions about the original map.
- Translate coordinate results back to the original vector space when necessary.

The Standard Basis of Some Common Vector Spaces

- $\mathbb{R}^n$:
  - $\mathcal{E} = (\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n)$, where $\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, $\vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$, …, $\vec{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$.
- $\mathbb{R}^{2\times 2}$:
  - $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$, where $E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.
- $\mathbb{R}_n[x]$: polynomials in the variable x of degree at most n
  - $\mathcal{E} = (1, x, x^2, \ldots, x^n)$.

How to get M

- Assume the map in the question is $f : V \longrightarrow V$, the basis of V is $\mathcal{E} = (\beta_1, \beta_2, \ldots, \beta_n)$, (in this course exam, usually domain $=$ codomain——for testing eigenvalue and eigenvectors, if domain $\neq$ codomain, the idea is similar).
- The coordinate matrix M of f with respect to the standard basis of both domain and codomain is:
  - $M = [[f(\beta_1)]_\mathcal{E}, [f(\beta_2)]_\mathcal{E}, \ldots, [f(\beta_n)]_\mathcal{E}]$
  - Example:
    - In the exam question, we've known: $A = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 2}$, map $f_A : \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ given by $f_A(B) = AB$
      - so $\mathcal{E} = (\beta_1, \beta_2, \ldots, \beta_n) = (E_{11}, E_{12}, E_{21}, E_{22})$,
        - where $E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.
    - So $M = [[f(\beta_1)]_\mathcal{E}, [f(\beta_2)]_\mathcal{E}, \ldots, [f(\beta_n)]_\mathcal{E}] = [[f(E_{11})]_\mathcal{E}, [f(E_{12})]_\mathcal{E}, [f(E_{21})]_\mathcal{E}, [f(E_{22})]_\mathcal{E}]$,
      - Then calculate $f(E_{11}) = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}$, $f(E_{12}) = \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}$, $f(E_{21}) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, $f(E_{22}) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.
      - Then calculate $[f(E_{11})]_\mathcal{E} = \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$; $[f(E_{12})]_\mathcal{E} = \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \end{pmatrix}$; $[f(E_{21})]_\mathcal{E} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$; $[f(E_{22})]_\mathcal{E} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$.
    - So we get $M = M_\mathcal{E}^\mathcal{E}(f_A) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$.
- Writing:
  - Let $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$ be the standard basis of $\mathbb{R}^{2\times 2}$, where $E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, and $E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.
  - The matrix associated with f_A with respect to $\mathcal{E}$ is
    $M = ([f_A(E_{11})]_\mathcal{E}, [f_A(E_{12})]_\mathcal{E}, [f_A(E_{21})]_\mathcal{E}, [f_A(E_{22})]_\mathcal{E})$
    $= \left(\left[\begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}\right]_\mathcal{E}, \left[\begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}\right]_\mathcal{E}, \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right]_\mathcal{E}, \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right]_\mathcal{E}\right)$
    $= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$.
- Figure:

  • Let $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$ be the standard basis of $\mathbb{R}^{2\times 2}$, where

  $$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

  • The matrix associated with f_A with respect to $\mathcal{E}$ is

  $$M = ([f_A(E_{11})]_\mathcal{E}, [f_A(E_{12})]_\mathcal{E}, [f_A(E_{21})]_\mathcal{E}, [f_A(E_{22})]_\mathcal{E})$$
  $$= \left(\left[\begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}\right]_\mathcal{E}, \left[\begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}\right]_\mathcal{E}, \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right]_\mathcal{E}, \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right]_\mathcal{E}\right)$$
  $$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

Solve the Following Questions

- We've already obtained the coordinate matrix M of the map f in the question, and then we can solve all the questions through M.
- For the validity of the method, see the section “What Properties are Preserved” above.

Endomorphism

- For this question, we don't need M.
- How to explain why f is an endomorphism:
  1. $f : V \longrightarrow V$, namely the domain and codomain are the same:
     - Usually it is obvious.
  2. f is a linear map:
     - Prove $f(B_1 + B_2) = f(B_1) + f(B_2)$ and $f(\lambda B_1) = \lambda f(B_1)$.
       - Example in the exam:
         - $f_A(B_1 + B_2) = A(B_1 + B_2) = AB_1 + AB_2 = f_A(B_1) + f_A(B_2)$,
         - $f_A(\lambda B_1) = A(\lambda B_1) = \lambda AB_1 = \lambda f_A(B_1)$.
     - Or prove $f(\lambda B_1 + \delta B_2) = \lambda f(B_1) + \delta f(B_2)$.
       - Example in the exam:
         - $f_A(\lambda B_1 + \delta B_2) = A(\lambda B_1 + \delta B_2) = \lambda AB_1 + \delta AB_2 = \lambda f_A(B_1) + \delta f_A(B_2)$.

$\mathrm{Im}(f)$

- $\mathrm{Im}(f) = \{(\vec{\beta}_1, \vec{\beta}_2, \ldots, \vec{\beta}_n)\vec{c} \mid \vec{c} \in [\mathrm{Im}(f)]_{\mathcal{E}}\} = \{(\vec{\beta}_1, \vec{\beta}_2, \ldots, \vec{\beta}_n)\vec{c} \mid \vec{c} \in \mathrm{Im}(M)\}$
- So, we can find a basis of $\mathrm{Im}(f)$ by finding a basis of $\mathrm{Col}(M)$ and then translating those coordinate vectors back to the original vector space.
- Example:
  - We've known: $A = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 2}$, map $f_A : \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ given by $f_A(B) = AB$
    - And we get $M = M_{\mathcal{E}}^{\mathcal{E}}(f_A) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$, $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$.
  - How to find $\mathrm{Im}(f)$:
    - First, find a basis of $\mathrm{Col}(M)$:
      - $M = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \xrightarrow{\text{Row reduction}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $\vec{m}_1, \ldots, \vec{m}_4$ are columns of M.
      - So, we find the $\mathrm{rank}(M) = 2$, and then we can find that a basis of $\mathrm{Im}(M)$ is $\{\begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \end{pmatrix}\}$.
        - Notice, the basis of $\mathrm{Im}(M)$ should not be the column vectors of the row-reduced matrix ~~$\{\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}\}$~~!
        - Instead, we have to take the pivot columns in the original matrix M.
- Second, translate the basis of $\mathrm{Im}(M)$ back to the basis of $\mathrm{Im}(f_A)$:
  - $\begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}$.
    - Computation:
    - $(E_{11}, E_{12}, E_{21}, E_{22}) \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}$, $(E_{11}, E_{12}, E_{21}, E_{22}) \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}$
- Therefore, a basis of $\mathrm{Im}(f_A)$ is $\{\begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}\}$.

Preimage $f^{-1}(\vec{x})$

- $f^{-1}(\vec{x}) = \{\vec{v} \in V | f(\vec{v}) = \vec{x}\}$, $[f(\vec{v})]_{\mathcal{E}} = M[\vec{v}]_{\mathcal{E}}$
- $[f^{-1}(\vec{x})]_{\mathcal{E}} = \{[\vec{v}]_{\mathcal{E}} | \vec{v} \in V, M[\vec{v}]_{\mathcal{E}} = [\vec{x}]_{\mathcal{E}}\}$, namely $[f^{-1}(\vec{x})]_{\mathcal{E}} = \{\vec{c} \in \mathbb{R}^n | M\vec{c} = [\vec{x}]_{\mathcal{E}}\}$
- Therefore, we can find the preimage of $\vec{x}$ under f by solving non-homogeneous system $M\vec{c} = [\vec{x}]_{\mathcal{E}}$ and then translating those coordinate vectors back to the original vector space.
  - For $M\vec{c} = [\vec{x}]_{\mathcal{E}}$:
    - If $\vec{x} \in \mathrm{Im}(f)$, equivalently $[\vec{x}]_{\mathcal{E}} \in \mathrm{Im}(M)$, then the system has solutions;
    - otherwise, the preimage is empty.
- Example:
  - We've known: $A = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 2}$, map $f_A : \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ given by $f_A(B) = AB$
    - And we get $M = M_{\mathcal{E}}^{\mathcal{E}}(f_A) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$, $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$.
    - A basis of $\mathrm{Im}(M)$ is $\{\begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \end{pmatrix}\}$.
    - A basis of $\mathrm{Im}(f_A)$ is $\{\begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}\}$.
- How to find $f_A^{-1}(\vec{x})$:
  - Assume $\vec{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $[\vec{x}]_{\mathcal{E}} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$.
  - First, solve $M\vec{c} = [\vec{x}]_{\mathcal{E}}$:
    - Augmented matrix is $(M \mid [\vec{x}]_{\mathcal{E}}) = \left(\begin{array}{cccc|c} -1 & 0 & 0 & 0 & a \\ 0 & -1 & 0 & 0 & b \\ -1 & 0 & 0 & 0 & c \\ 0 & -1 & 0 & 0 & d \end{array}\right) \xrightarrow{\text{Row reduction}} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -a \\ 0 & 1 & 0 & 0 & -b \\ 0 & 0 & 0 & 0 & a-c \\ 0 & 0 & 0 & 0 & b-d \end{array}\right)$
    .
    - This system has solutions iff $a = c$ and $b = d$, so $\vec{x} = \begin{pmatrix} a & b \\ a & b \end{pmatrix}$, $[\vec{x}]_{\mathcal{E}} = \begin{pmatrix} a \\ b \\ a \\ b \end{pmatrix}$.
  - So $(M \mid [\vec{x}]_{\mathcal{E}}) = \left(\begin{array}{cccc|c} -1 & 0 & 0 & 0 & a \\ 0 & -1 & 0 & 0 & b \\ -1 & 0 & 0 & 0 & a \\ 0 & -1 & 0 & 0 & b \end{array}\right) \xrightarrow{\text{Row reduction}} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -a \\ 0 & 1 & 0 & 0 & -b \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$
  - Therefore, a particular solution of $M\vec{c} = [\vec{x}]_{\mathcal{E}}$ is $\begin{pmatrix} -a \\ -b \\ 0 \\ 0 \end{pmatrix}$.
  - And a basis of $\mathrm{Ker}(M)$ is $\{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\}$.

- Hence the solution of $M\vec{c} = [\vec{x}]_{\mathcal{E}}$ is $\{s \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -a \\ -b \\ 0 \\ 0 \end{pmatrix} | s, t \in \mathbb{R}\} = \{ \begin{pmatrix} -a \\ -b \\ s \\ t \end{pmatrix} | s, t \in \mathbb{R}\}$,

  where $[\vec{x}]_{\mathcal{E}} = \begin{pmatrix} a \\ b \\ a \\ b \end{pmatrix}$.

- Second, translate the solution of $M\vec{c} = [\vec{x}]_{\mathcal{E}}$ back to original vector space:
  - $\begin{pmatrix} -a \\ -b \\ s \\ t \end{pmatrix} \leftrightarrow \begin{pmatrix} -a & -b \\ s & t \end{pmatrix}$
    - Computation:
      - $(E_{11}, E_{12}, E_{21}, E_{22}) \begin{pmatrix} -a \\ -b \\ s \\ t \end{pmatrix} = \begin{pmatrix} -a & -b \\ s & t \end{pmatrix}$.
    - Therefore, $f_A^{-1}(\vec{x}) = \{\begin{pmatrix} -a & -b \\ s & t \end{pmatrix} | s, t \in \mathbb{R}\}$, where $\vec{x} = \begin{pmatrix} a & b \\ a & b \end{pmatrix}$.

- How to find $\vec{x}$ such that $f_A^{-1}(\vec{x}) = \varnothing$: $f_A^{-1}(\vec{x}) = \varnothing \iff \vec{x} \notin \mathrm{Im}(f_A)$
  - $f_A^{-1}(\vec{x}) = \varnothing$ means $M\vec{c} = [\vec{x}]_{\mathcal{E}}$ has no solution, and we know augmented matrix is

    $$(M \mid [\vec{x}]_{\mathcal{E}}) = \left(\begin{array}{cccc|c} -1 & 0 & 0 & 0 & a \\ 0 & -1 & 0 & 0 & b \\ -1 & 0 & 0 & 0 & c \\ 0 & -1 & 0 & 0 & d \end{array}\right) \xrightarrow{\text{Row reduction}} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -a \\ 0 & 1 & 0 & 0 & -b \\ 0 & 0 & 0 & 0 & a-c \\ 0 & 0 & 0 & 0 & b-d \end{array}\right).$$

  - Therefore, when $a \neq c$ or $b \neq d$, $f_A^{-1}(\vec{x}) = \varnothing$.
  - Hence for $f_A^{-1}(\vec{x}) = \varnothing$, $\vec{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a \neq c$ or $b \neq d$.
    - Namely, $\{\vec{x} \in \mathbb{R}^{2\times 2} \mid f_A^{-1}(\vec{x}) = \varnothing\} = \{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2\times 2} \mid a \neq c \text{ or } b \neq d\}$.

$\mathrm{Ker}(f)$

- $\mathrm{Ker}(f) = \{\vec{x} \in V | f(\vec{x}) = \vec{0}\} = \{\vec{x} = (\vec{\beta}_1, \vec{\beta}_2, \ldots, \vec{\beta}_n)\vec{c} = c_1\vec{\beta}_1 + c_2\vec{\beta}_2 + \cdots + c_n\vec{\beta}_n | M\vec{c} = \vec{0}\}$
- So, we can find $\mathrm{Ker}(f)$ by finding $\mathrm{Ker}(M)$ and then translating those coordinate vectors back to the original vector space.
- Example:
  - We've known: $A = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 2}$, map $f_A : \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ given by $f_A(B) = AB$
    - And we get $M = M_{\mathcal{E}}^{\mathcal{E}}(f_A) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$, $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$.
  - How to find $\mathrm{Ker}(f_A)$:
    - First, solve homogeneous system $M\vec{c} = \vec{0}$ to find $\mathrm{Ker}(M)$:
      - $M = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \xrightarrow{\text{Row reduction}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.
    - Therefore, we can find that a basis of $\mathrm{Ker}(M)$ is $\{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\}$.
- Second, translate the basis of $\mathrm{Ker}(M)$ back to the basis of $\mathrm{Ker}(f_A)$:
  - $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.
    - Computation:
      - $(E_{11}, E_{12}, E_{21}, E_{22}) \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, (E_{11}, E_{12}, E_{21}, E_{22}) \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$
  - Therefore, a basis of $\mathrm{Ker}(f_A)$ is $\{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\}$.

Eigenvalues and Eigenvectors

- λ is an eigenvalue of f iff λ is an eigenvalue of M.
  - $\mathrm{Spec}(f)$:
    - The spectrum of a linear map $f : V \to V$, denoted by $\mathrm{Spec}(f)$, is the set of all eigenvalues of f.
    - Equivalently,
      - $\mathrm{Spec}(f) = \{\lambda \in \mathbb{R} \mid \exists \vec{v} \in V,\ \vec{v} \neq \vec{0},\ f(\vec{v}) = \lambda \vec{v}\}$
    - In terms of the matrix representation,
      - $\mathrm{Spec}(f) = \mathrm{Spec}(M) = \{\lambda \in \mathbb{R} \mid \det(M - \lambda I) = 0\}$
- Characteristic polynomial: $p_f(\lambda) := p_M(\lambda) = \det(M - \lambda I)$
- The eigenvectors of M are the representations of the eigenvectors of f:
  - $E_\lambda(f) = \{\vec{x} \in V | f(\vec{x}) = \lambda \vec{x}\}$,
  - $[E_\lambda(f)]_\mathcal{E} = E_\lambda(M) = \{[\vec{x}]_\mathcal{E} \in \mathbb{R}^n | M[\vec{x}]_\mathcal{E} = \lambda [\vec{x}]_\mathcal{E}\} = \mathrm{Ker}(M - \lambda I)$
- Diagonal form:
  - A diagonal form of a linear map $f : V \to V$ is a diagonal matrix representation of f with respect to a basis of V consisting of eigenvectors of f.
    - If such a basis exists, then f is diagonalizable.
    - $[f]_\mathcal{B} = D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, where $\mathcal{B} = (\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n)$ is a basis of eigenvectors and $f(\vec{v}_i) = \lambda_i \vec{v}_i$.
- Example:
  - We've known: $A = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 2}$, map $f_A : \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ given by $f_A(B) = AB$
    - And we get $M = M_\mathcal{E}^\mathcal{E}(f_A) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$, $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$.
  - How to find eigenvalues and eigenvectors of f_A:
    - First, find eigenvalues by calculating the characteristic polynomial of M:
      - $p_{f_A}(\lambda) = p_M(\lambda) = \det(M - \lambda I) = \lambda^2(\lambda + 1)^2$,
      - Therefore, $\lambda = 0$, $\lambda = -1$, $m_a(0) = 2$, $m_a(-1) = 2$
    - Second, find the bases of eigenspaces of M:
      - For $\lambda = 0$, the eigenspace of M is $\mathrm{Ker}(M)$:
        - So a basis of $E_0(M)$ is $\{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\}$.
      - For $\lambda = -1$, the eigenspace of M is $\mathrm{Ker}(M - \lambda I_4)$,
        - $M - \lambda I_4 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} \xrightarrow{\text{Row reduction}} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.
        - So a basis of $E_{-1}(M)$ is $\{\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}\}$.
    - Third, translate the basis of the eigenspace of M back to the basis of the eigenspace of f_A:
      - For $\lambda = 0$, the eigenspace of f_A is $\mathrm{Ker}(f_A)$.
        - A basis of $E_0(f_A)$ is $\{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\}$.

        - $E_0(f_A)$ is $\{s_1 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + t_1 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} | s_1, t_1 \in \mathbb{R}\}$.
    - For $\lambda = -1$, the eigenspace of f_A is $\mathrm{Ker}(f_A - \lambda \, \mathrm{id}_{\mathbb{R}^{2\times 2}})$:
        - $\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$.
            - Computation:
                - $(E_{11}, E_{12}, E_{21}, E_{22}) \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, (E_{11}, E_{12}, E_{21}, E_{22}) \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$
        - So, a basis of $E_{-1}(f_A)$ is $\{\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}\}$.
        - $E_{-1}(f_A)$ is $\{s_2 \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} + t_2 \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} | s_2, t_2 \in \mathbb{R}\}$.
- Diagonal form of M:
    - $D = \mathrm{diag}(0, 0, -1, -1)$, where $\mathcal{C} = \left(\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right)$ is an eigenbasis of M.
- Diagonal form of f_A:
    - $D = \mathrm{diag}(0, 0, -1, -1)$, where $\mathcal{B} = \left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right)$ is an eigenbasis of f_A.